

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering **ASSESSMENT TOOL 3 – Comparative Analysis**

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO4 – Articulation (combining methods for evaluation) P4	Assessment:	Assessment Tool – Comparative Analysis
Weightage:	10% of CIA	Total Marks:	100
CO4 Statement:	Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL5)		
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Section / Batch:	Third Year/ B.Tech IT	Date:	25/08/2026

Code of Conduct

I Kaviya V/192421022 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Kaviya V

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant	
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison	
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided	
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation	
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

QUESTIONS:

1. Detection Performance Comparison

Two object detection models produce the following results:

Model True Positives False Positives False Negatives

A (Viola-Jones) 80 20 30

B (YOLO) 90 15 20

(a) Compute precision and recall for both models.

(b) Compare their performance and state which model is better for object detection.

2. Motion Estimation Error Analysis

Two motion estimation techniques produce displacement errors (in pixels):

Frame Method A Method B

1 2 3

2 3 5

3 2 4

(a) Compute average error for both methods.

(b) Compare and evaluate which method is more accurate.

3. Optical Flow Magnitude Comparison

Optical flow magnitudes (in pixels/frame):

Region Method A Method B

R1 5 8

R2 6 10

R3 5 9

(a) Compute average motion magnitude for both methods.

(b) Compare and evaluate which method captures stronger motion.

ANSWERS:

1. Detection Performance Comparison

Given Data:

Model	True Positives (TP)	False Positives (FP)	False Negatives (FN)
A – Viola-Jones	80	20	30
B – YOLO	90	15	20

(a) Precision and Recall

Formulas

Precision: $\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$

Recall: $\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$

Model A – Viola-Jones:

Precision:

$$\begin{aligned} &= 80/(80+20) \\ &= 80/100 = 0.80 \end{aligned}$$

Precision = 80%

Recall:

$$\begin{aligned} &= 80/(80+30) \\ &= 80/110 = 0.727 \end{aligned}$$

Recall $\approx$ 72.73%

Model B – YOLO:

Precision:

$$\begin{aligned} &= 90/(90+15) = 90/105 \\ &= 0.857 \end{aligned}$$

Precision $\approx$ 85.71%

Recall:

$$\begin{aligned} &= 90/(90+20) \\ &= 90/110 = 0.818 \end{aligned}$$

Recall $\approx$ 81.82%

(b) Comparison

Model	Precision	Recall
Viola-Jones	80%	72.73%
YOLO	85.71%	81.82%

Conclusion:

YOLO is better for object detection because it has both higher precision and higher recall than Viola-Jones.

- YOLO has fewer false positives.
- YOLO has fewer false negatives.
- YOLO detects more actual objects correctly.

Best Model: YOLO

2. Motion Estimation Error Analysis

Given Data:

Frame	Method A	Method B
1	2	3
2	3	5
3	2	4

(a) Average Error:

Method A:

$$\begin{aligned}\text{Average Error} &= (2+3+2)/3 \\ &= 7/3 = 2.33 \text{ pixels}\end{aligned}$$

Method B:

$$\begin{aligned}\text{Average Error} &= (3+5+4)/3 \\ &= 12/3 = 4 \text{ pixels}\end{aligned}$$

(b) Comparison:

Method	Average Error
A	2.33 pixels
B	4 pixels

Conclusion:

Method A is more accurate because it has a lower average displacement error.

$$2.33 < 4$$

A lower error indicates that the estimated motion is closer to the actual displacement.

Best Method: A

3. Optical Flow Magnitude Comparison

Given Data:

Region	Method A	Method B
R1	5	8
R2	6	10
R3	5	9

(a) Average Motion Magnitude:

Method A:

$$\begin{aligned}\text{Average} &= (5+6+5)/3 \\ &= 16/3 = 5.33 \text{ pixels/frame}\end{aligned}$$

Method B:

$$\begin{aligned}\text{Average} &= (8+10+9)/3 \\ &= 27/3 = 9 \text{ pixels/frame}\end{aligned}$$

(b) Comparison:

Method	Average Motion Magnitude
A	5.33 pixels/frame
B	9 pixels/frame

Conclusion:

Method B captures stronger motion because its average optical-flow magnitude is higher:

$$9 > 5.33$$

Therefore, Method B indicates greater estimated motion magnitude across the three regions.

Best Method for stronger motion capture: B